

(1001CJA102120017)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

UNIT TEST

21-06-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)**ANSWER KEY****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,D	A,B,D	A,C	B,C	B,C	C	B	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	20.00	1.00	3.00	60.00	0.40	6.28 to 6.29	1.00	6.40		

PART-2 : CHEMISTRY

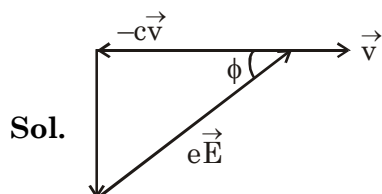
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,C,D	A,B,C,D	A,D	B,D	B,D	B	B	A	D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	11.40 to 11.50	57.40 to 60.00	8.00	8.00	-1.50	0.66 or 0.67	2.50	1.33 or 1.34		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A,B,C	C,D	B,D	B,C	A,B,C	C	B	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	3.14	-0.14	1.08	1.00	2.00	0.25	-24.00	2.00		

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)**PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (A,C)



$$e\vec{v} \times \vec{B}$$

$\vec{B}$ is $\perp$ to this plane as well as E

$$\Rightarrow \vec{B} \perp \text{to } \vec{v}$$

$$eE = cv_0$$

$$eE \cos \phi = cv$$

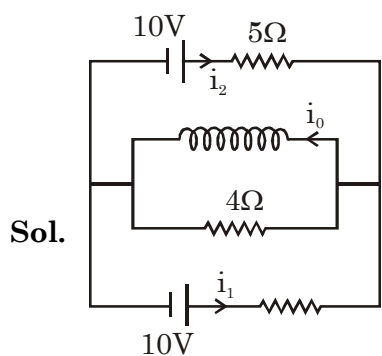
$$eE \sin \phi = evB$$

$$\cos \phi = \frac{v}{v_0}$$

$$E \sin \phi = v_0 \cos \phi B$$

$$\tan \phi = \frac{Bv_0}{E}$$

2. Ans. (A,D)



$$i_0 = \frac{10}{20} = 0.5 \text{ A}$$

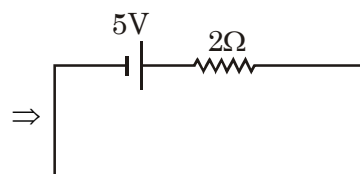
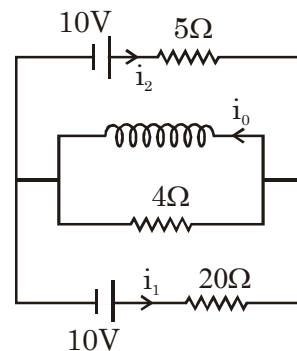
$$v_{eq} = \frac{\frac{10}{20} + \frac{10}{5} + \frac{0}{4}}{\frac{1}{20} + \frac{1}{5} + \frac{1}{4}} = \frac{2.5}{\frac{1+4+5}{20}} = 5$$

$$10 - 20i_1 = 5 \Rightarrow i_1 = 0.25 \text{ A}$$

$$10 - 5i_2 = 5 \Rightarrow i_2 = 1 \text{ A}$$

$$i_3 = 1.25 - 0.5 = 0.75 \text{ A}$$

After long time



$$0 = 10 - 20i_2$$

$$i_1 = 0.5 \text{ A}$$

3. Ans. (A,B,D)

Sol. $2\pi r = 314$

$$r_1 = 50 \text{ cm}$$

$$M_1 = \pi r^2 \times i$$

$$= 3 \times 157 \times 50$$

$$= 150 \times 157$$

$$4\ell = 314$$

$$\ell = \frac{157}{2}$$

$$M_2 = \ell^2 i$$

$$= \frac{157^2}{4} \times 3$$

$$3\ell' = 314$$

$$M_3 = \frac{\sqrt{3}\ell'^2}{4} \times i$$

$$= \frac{3\sqrt{3}}{4} \times \frac{314^2}{9} = \frac{157^2}{\sqrt{3}}$$

$$M_4 = 10 \times \pi r^2 \times i$$

$$2\pi r \times 10 = 314$$

$$M_4 = 10 \times \pi \times 25 \times 3$$

4. Ans. (A,C)

Sol. $\frac{kq}{r} = \frac{V}{2}k$ & $\frac{kq}{r} = \frac{V}{2}$

$$\text{Total charge flows} = q - q' = \frac{r}{k} V$$

$$d = \frac{1}{2} \times q \times \frac{V}{md} t^2$$

$$\frac{2md^2}{2\pi\epsilon_0 r V} = t^2 \Rightarrow \frac{\sqrt{md}}{\sqrt{\pi\epsilon_0 r V}}$$

$$i = \frac{2(\pi\epsilon_0 r)^{3/2} V^2}{\sqrt{md}}$$

$$P = \frac{\Delta U}{t} = \frac{qV}{t} = \frac{2\pi\epsilon_0 r V^2}{\sqrt{\pi m m r}}$$

5. Ans. (B, C)

Sol. Magnetic flux through the smaller loop due to current in the bigger loop

$$= \left[\frac{\mu_0 I}{2\pi d} - \frac{\mu_0 I}{2\pi(2d)} \right] \ell^2 = \frac{\mu_0 I \ell^2}{4\pi d}$$

$$\text{So mutual inductance} = \frac{\mu_0 \ell^2}{4\pi d}$$

So emf induced in the bigger loop

$$= \frac{d}{dt} \left[\frac{\mu_0 \ell^2}{4\pi d} k t \right] = \frac{\mu_0 \ell^2 k}{4\pi d}$$

6. Ans. (B, C)

$$\text{Sol. } \varepsilon = -\frac{d\phi}{dt} = B\ell v$$

$$Q = \frac{\Delta\phi}{R} = \text{independent of } v$$

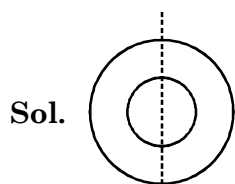
$$\text{area} = \Delta\phi$$

7. Ans. (C)

$$\text{Sol. } E \times 4\pi R^2 = \frac{\rho_0 \times \frac{4}{3} \pi R_S^3 - \rho \left(\frac{4}{3} \pi R^3 - \frac{4}{3} \pi R_S^3 \right)}{\epsilon_0}$$

$$E = \frac{\rho_0}{3\epsilon_0} \left(\frac{2R_S^3}{R^2} - R \right)$$

8. Ans. (B)



$$\vec{M} = \vec{M}_1 + \vec{M}_2$$

$$\vec{M}_1 = \frac{Q}{2m} I \omega = \frac{\rho_0 \times \frac{4}{3} \pi R_S^3}{2m} \times \frac{2}{5} m R_S^2 \omega$$

$$\vec{M}_2 = \frac{-\rho_0 \times \frac{4}{3} \pi (R^3 - R_S^3)}{2\sigma \times \frac{4}{3} \pi (R^3 - R_S^3)}$$

$$\times \omega \int_{R_S}^R \frac{2}{3} x^2 \times 4\pi x^2 \sigma dx$$

$$\vec{M}_1 = \frac{4\pi}{15} \rho_0 R_S^5 \vec{\omega}$$

$$\vec{M}_2 = \frac{-\rho_0 \omega}{20} \times \frac{2}{3} \times 4\pi \frac{(R^5 - R_S^5)}{5}$$

$$= \frac{4\pi \rho_0 \omega}{15} (R^5 - R_S^5)$$

$$\vec{M}_1 + \vec{M}_2 = 0$$

$$\Rightarrow 2R_S^5 - R^5 = 0$$

$$R_S = \frac{R}{2^{1/5}}$$

9. Ans. (B)

10. Ans. (B)

Sol. Loss of P.E. of rod

$$\frac{1}{2} Li^2 + \frac{1}{2} mv^2$$

$$mgh = \frac{1}{2} Li^2 + \frac{1}{2} mv^2 \quad \dots (i)$$

When v is maximum $F_{\text{net}} = 0$

$$i_1 \ell B = mg$$

$$i_1 = mg/\ell B$$

$$\text{Now maximum current } (i_{\text{max}}) = \frac{2mg}{\ell B}$$

When i_{max} then $v = 0$ then

$$\frac{1}{2} L \left(\frac{2mg}{\ell B} \right)^2 = mgx_{\text{max}}$$

$$x_{\text{max}} = \frac{2g}{B^2 \ell^2} mL$$

$$\text{When v is maximum : } h = \frac{x_{\text{max}}}{2} \text{ and } i = i_1$$

$$\text{put in (i) : } mg \frac{\left(\frac{2gmL}{B^2 \ell^2} \right)}{2} = \frac{1}{2} L \left(\frac{mg}{\ell B} \right)^2 + \frac{1}{2} mv_{\text{max}}^2$$

$$L \left(\frac{mg}{\ell B} \right)^2 = mv_{\text{max}}^2$$

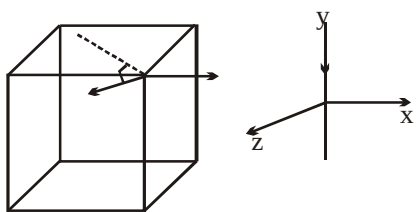
$$v_{\text{max}} = \frac{g}{B\ell} \sqrt{mL}$$

SECTION-II

1. Ans. 20.00 V

Sol. $\frac{V^2}{\rho \ell} \times \pi r^2 = \frac{dH}{dt} = 2\pi r \ell \times 4T_0^3 \times \sigma(T - T_s)$
 $\Rightarrow V^2 \propto \ell^2$
 $\Rightarrow V \propto \ell$

2. Ans. 1.00

Sol. 

$$B = \frac{4\pi \times 10^{-7}}{2\pi} \times \frac{8}{\sqrt{2}} \left[\frac{1}{\sqrt{2}} \hat{i} + \frac{1}{\sqrt{2}} \hat{k} \right] + 1.4 \times 10^{-6} \hat{i}$$

$$= (-8\hat{i} + 8\hat{k} + 14\hat{i}) \times 10^{-7}$$

$$= (6\hat{i} + 8\hat{k}) \times 10^{-7}$$

$$B = 10 \times 10^{-7} \text{ T}$$

$$\text{Ans } 1 \mu\text{T}$$

3. Ans. 3.00

Sol. $E \times 4\pi x^2 = \frac{\int_0^x k r^{-n} \times 4\pi r^2 dr}{t_0} = \frac{4\pi k x}{\epsilon_0(3-x)}$

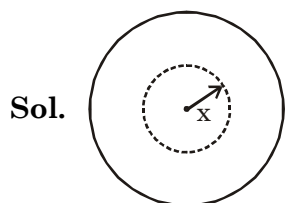
$$E = \frac{4\pi k x^{1-n}}{\epsilon_0(3-x)}$$

$$v = \int \frac{1}{2} E_0 E^2 dV = \infty$$

$$\Rightarrow 3-x=0$$

$$n=3$$

4. Ans. 60.00

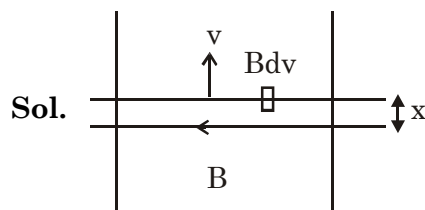


$$\phi = B_0 t \int_0^r \left(1 - \frac{r}{R}\right) \times 2\pi r dr = B_0 t 2\pi \left(\frac{r^2}{2} - \frac{r^3}{3R}\right)$$

$$E \times 2\pi r = \frac{d\phi}{dt} = B_0 2\pi \left(\frac{r^2}{2} - \frac{r^3}{3R}\right)$$

$$\frac{dE}{dr} = 0 \Rightarrow \frac{1}{2} - \frac{2r}{3} \Rightarrow r = \frac{3r}{4}$$

5. Ans. 0.40



$$i = \frac{Bdv}{2\rho d + 2\rho x} \approx \frac{Bv}{2\rho}$$

$$F = idB - \frac{\mu_0 i^2}{2\pi x} d$$

$$\Rightarrow B = \frac{\mu_0 i}{2\pi x}$$

$$x = \frac{\mu_0 Bv}{4\pi \rho B}$$

6. Ans. 6.28 to 6.29

Sol. $900 = \frac{9 \times 10^9 \times Q}{10^{-2}}$

$$Q = 1 \text{ nC}$$

$$B = \frac{\mu_0 qv}{4\pi r^2} = \frac{10^{-7} \times 10^{-9} \times 2\pi \times 300 \times 0.3}{(0.3)^2}$$

$$= 10^{-14} \times \frac{6\pi}{3} = 6.28 \times 10^{-14} \text{ T}$$

7. Ans. 1.00

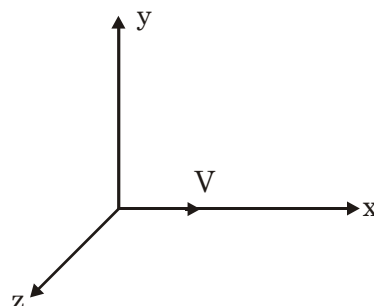
Sol. $\Delta V = Ed = \frac{Q}{C} \Rightarrow Q = CEd$

$$i_{\max} = CEd\omega$$

$$= 10^{-3} \times 10^3 \times 10^{-3} \times \frac{1}{\sqrt{10^{-3} \times 10^{-3}}} = 1 \text{ A}$$

8. Ans. 6.40

Sol. Magnetic force in $\hat{j}$ direction and electric field is in $-\hat{j}$ direction



$$\text{Resultant force} = qVB - qE$$

$$= q(1.28 \times 10^5 \times 8 \times 10^{-2} - 102.4 \times 10^3) = 0$$

$$R = 100 \text{ m}$$

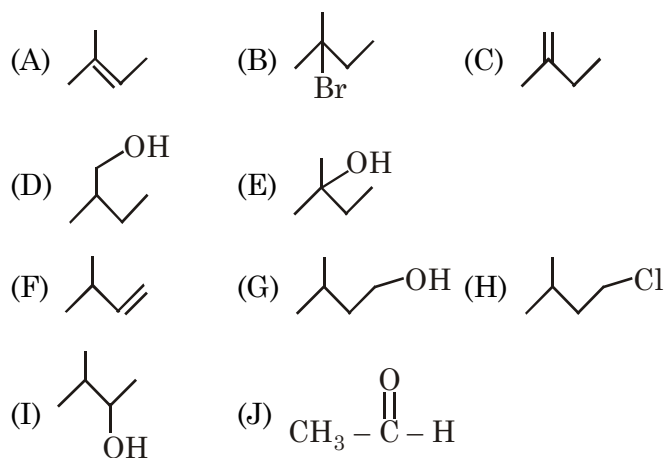
Charge will move only in x-direction.

$$x = V \times t = 1.28 \times 10^6 \times 5 \times 10^{-6} = 6.4 \text{ m}$$

PART-2 : CHEMISTRY**SOLUTION****SECTION-I**

1. Ans.(A,C)
2. Ans.(A,C,D)
3. Ans.(A,B,C,D)
4. Ans.(A,D)
5. Ans.(B,D)
6. Ans.(B,D)
7. Ans.(B)
8. Ans.(B)

Sol. for Question number 7 and 8



9. Ans.(A)
10. Ans.(D)

SECTION-II

1. Ans.(11.40 to 11.50)

$$\Delta G = nRT \ln \frac{P_2}{P_1}$$

$$= 2 \times 8.3 \times 300 \ln 10$$

$$= 2 \times \frac{8.3}{1000} \times 300 \times 2.303 = 11.47 \text{ kJ}$$

2. Ans.(57.40 to 60.00)

$$\pi = \frac{\left(\frac{1}{2} + \frac{2}{3} \times 2\right) \times 0.08 \times 300}{\left(\frac{932}{466/375} \times 10^{-3}\right)} = 58.56 \text{ atm}$$

3. Ans.(8.00)
4. Ans.(8.00)
5. Ans.(-1.50)
6. Ans.(0.66 or 0.67)
7. Ans.(2.50)
8. Ans.(1.33 or 1.34)

PART-3 : MATHEMATICS**SOLUTION****SECTION-I**

1. Ans. (C)

Sol. $\int \frac{1}{\sqrt{2} + \sqrt{2} \sin\left(x + \frac{\pi}{4}\right)} dx = \frac{1}{\sqrt{2}} \int \frac{1 - \sin\left(x + \frac{\pi}{4}\right)}{\cos^2\left(x + \frac{\pi}{4}\right)} dx$

$$\Rightarrow \frac{1}{\sqrt{2}} \left[\int \sec^2\left(x + \frac{\pi}{4}\right) - \int \sec\left(x + \frac{\pi}{4}\right) \tan\left(x + \frac{\pi}{4}\right) dx \right]$$

$$\Rightarrow \frac{1}{\sqrt{2}} \left[\tan\left(x + \frac{\pi}{4}\right) - \sec\left(x + \frac{\pi}{4}\right) \right] + C$$

2. Ans. (A,B,C)

Sol. $S = 4 \left[\int_0^{\pi/8} e^{\cos 4x} dx + \int_{\pi/8}^{\pi/4} e^{-\cos 4x} dx \right]$

Put $(4x) = t$,

$$S = 2 \int_0^{\pi/2} e^{\sin t} dt$$

Also, $2 \int_0^{\pi/2} 1 dt < 2 \int_0^{\pi/2} e^{\cos t} dt$

$$< 2 \int_0^{\pi/2} e^t dt$$

$$\Rightarrow \pi < S < 2(e^{\pi/2} - 1)$$

3. Ans. (C,D)

Sol. $|A^5 - 2A^4| = 0$ and $A^{-1} = \frac{1}{2}(A^2 - 5A + 7I)$

4. Ans. (B,D)

Sol. Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$|\vec{r}| = \sqrt{2} \Rightarrow x^2 + y^2 + z^2 = 2 \quad \dots(1)$$

$$\vec{p} \cdot \vec{q} = \vec{q} \cdot \vec{r} = \vec{r} \cdot \vec{p}$$

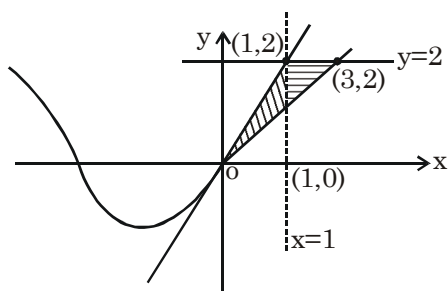
$$\Rightarrow 1 = y + z = x + y \quad \dots(2)$$

from (1) & (2)

$$\vec{r} = \hat{i} + \hat{k} \text{ and } \vec{r} = -\frac{1}{3}\hat{i} + \frac{4}{3}\hat{j} - \frac{1}{3}\hat{k}$$

5. Ans. (B,C)

Sol.



$$A_1 = \int_0^1 \left(\frac{(2x+1)^2 - 1}{4} - x \right) dx$$

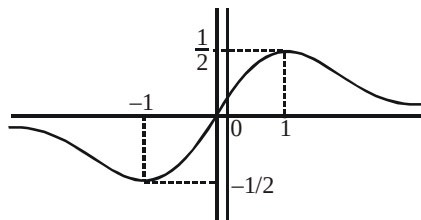
$$A_1 = \frac{1}{3}$$

$$A_2 = \frac{1}{2}$$

6. Ans. (A,B,C)

Sol. $\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{1}{1+x^2} \Rightarrow \frac{d}{dx}(1+x^2)y = 1$

$$\Rightarrow f(x) = \frac{x}{x^2+1}$$



Paragraph for Question 7 and 8

7. Ans. (C)

Sol. a $\rightarrow$ two entry

b $\rightarrow$ four entry

c $\rightarrow$ three entry

Number of c's in diagonal is either 1 or 3

$$= {}^3C_1 \left(\frac{|3|}{|2|} + \frac{|3|}{|2|} \right) + \frac{|3|}{|2|} = 30$$

8. Ans. (B)

Sol. Number of matrices A = 6

Paragraph for Question 9 & 10

Sol. $\frac{dy}{dx} - y = e^x g'(x)$

$$\Rightarrow ye^{-x} = \int g'(x) dx$$

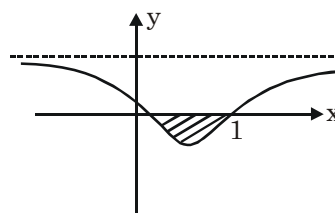
$$\Rightarrow ye^{-x} = \frac{x(x-1)}{x^2+1} e^{-x} + c$$

$$y = \frac{x(x-1)}{x^2+1} + ce^x \Rightarrow c = 0 \because f(0) = 0$$

$$\Rightarrow f(x) = \frac{x(x-1)}{(x^2+1)}$$

$$\frac{dy}{dx} = \frac{x^2+2x-1}{(x^2+1)^2} \Rightarrow y'(1) = \frac{1}{2}$$

$$\text{Area} \int_0^1 \frac{x^2-x}{x^2+1} dx$$



$$\int_0^1 \left(1 - \frac{x}{x^2+1} - \frac{1}{x^2+1} \right) dx$$

$$= 1 - \frac{1}{2} \ln 2 - \frac{\pi}{4} = \ln \left(\frac{e}{\sqrt{2}} \right) - \frac{\pi}{4}$$

9. Ans. (B)

10. Ans. (C)

SECTION-II

1. Ans. 3.14

Sol. $I = \int_1^3 \left[\tan^{-1} \left(\frac{(x-2)^2 - (x-2)}{(x-2)^3 + 1} \right) + \tan^{-1} \left(\frac{1}{(x-2)^2} \right) \right] dx$

$$= \int_{-1}^1 \left(\tan^{-1} \left(\frac{t^2-t}{t^3+1} \right) + \tan^{-1} \frac{1}{t^2} \right) dt$$

$$= \int_{-1}^1 \left(\tan^{-1} t^2 - \tan^{-1} t + \tan^{-1} \frac{1}{t^2} \right) dt$$

$$= \int_{-1}^1 \left(\tan^{-1} t^2 - \tan^{-1} t + \cot^{-1} t^2 \right) dt = \pi$$

2. Ans. -0.14

Sol. $I_n = \int \tan^{n-2} x \cdot \tan^2 x dx$

$$= \int \tan^{n-2} x \cdot (\sec^2 x - 1) dx$$

$$= \int \tan^{n-2} x \cdot \sec^2 x dx - \int \tan^{n-2} x dx$$

$$= \frac{(\tan x)^{n-1}}{n-1} - I_{n-2}$$

$$I_{n-2} + I_n = \frac{(\tan x)^{n-1}}{n-1}$$

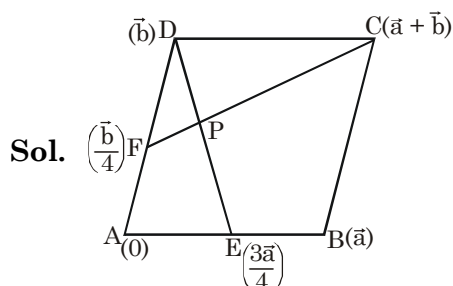
similarly

$$I_{n-4} + I_{n-2} = \frac{(\tan x)^{n-3}}{n-3}$$

$$I_n - I_{n-4} = \frac{(\tan x)^{n-1}}{n-1} - \frac{(\tan x)^{n-3}}{n-3}$$

now put $n = 10$

3. **Ans. 1.08**



Equation of ED is $\vec{r} = \vec{b} + \lambda(3\vec{a} - 4\vec{b})$

Equation of CF is $\vec{r} = \vec{a} + \vec{b} + \mu(4\vec{a} + 3\vec{b})$

$$\text{for P } \lambda = \frac{3}{25} \Rightarrow \vec{p} = \frac{13\vec{b} + 9\vec{a}}{25}$$

4. **Ans. 1.00**

Sol: $f'(x) = 1 + x \left[2x + \frac{2}{3}4x^3 + \frac{2}{3} \frac{4}{5}6x^5 + \dots \right]$

$$= 1 + x \frac{d}{dx}(xf(x)) = 1 + xf(x) + x^2 f'(x)$$

$$(1 - x^2)f'(x) = 1 + xf(x)$$

$$f(x) = \frac{\sin^{-1} x}{\sqrt{1-x^2}}, \text{ Now solve}$$

5. **Ans. 2.00**

Sol. $xdy - ydx = \frac{x^4}{y} dx$

$$\frac{y}{x} \left(\frac{xdy - ydx}{x^2} \right) = xdx$$

$$\Rightarrow \int \frac{y}{x} d\left(\frac{y}{x}\right) = \int xdx$$

$$\left(\frac{y}{x}\right)^2 = x^2 + 2c$$

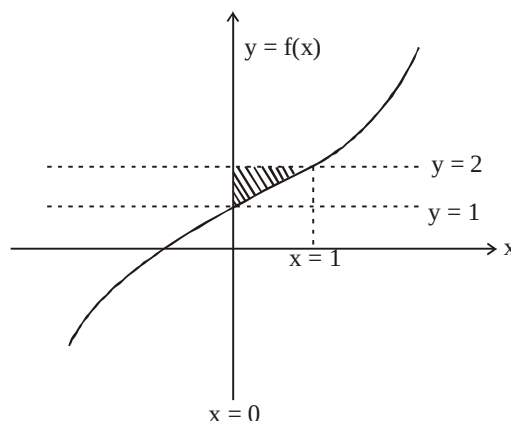
$$\Rightarrow y = \sqrt{x^4 + 2cx^2}$$

$$\text{put } x = 1 \text{ to get } c = \frac{3}{2}$$

$$\therefore f(x) = \sqrt{x^4 + 3x^2}$$

$$f(-1) = 2$$

6. **Ans. 0.25**



$$A = \int_0^1 2 - (x^3 - 3x^2 + 3x + 1) dx$$

$$A = \int_0^1 (-x^3 + 3x^2 - 3x + 1) dx$$

$$A = -\frac{x^4}{4} + x^3 - \frac{3x^2}{2} + x \Big|_0^1 = \frac{1}{4}$$

7. **Ans. -24.00**

$$A - B = \begin{bmatrix} 2 & 5 \\ 7 & 6 \end{bmatrix}$$

$$A^T - B^T = \begin{bmatrix} 2 & 7 \\ 5 & 6 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 2 & 7 \\ 5 & 6 \end{bmatrix}$$

$$2A = \begin{bmatrix} 4 & 12 \\ 12 & 12 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 6 \\ 6 & 6 \end{bmatrix}$$

$$|A| = -24$$

8. **Ans. 2.00**

Sol. Let position vectors of points A, B, C are $\vec{a}, \vec{b}, \vec{c}$ with respect to point P as origin

$\therefore$ Given condition

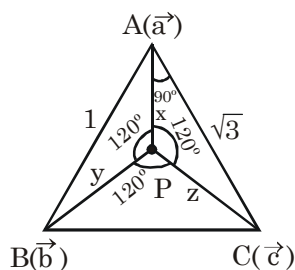
$$\Rightarrow \hat{a} + \hat{b} + \hat{c} = \vec{0} \text{ and } |\hat{a}| = |\hat{b}| \Rightarrow |\hat{c}| = 1$$

$$\Rightarrow \hat{a} + \hat{b} = -\hat{c} \text{ take mod and square}$$

$$|\hat{a} + \hat{b}|^2 = |\hat{c}|^2 \Rightarrow \cos \angle APB = -\frac{1}{2} \Rightarrow \angle APB = 120^\circ$$

$$\text{similarly } \angle APC = \angle BPC = 120^\circ$$

let $|\overrightarrow{PA}| = x, |\overrightarrow{PB}| = y$ and $|\overrightarrow{PC}| = z$



apply cosine rule in $\Delta APB, \Delta BPC, \Delta CPA$

$$x^2 + y^2 + xy = 1^2 \quad \dots(1)$$

$$y^2 + z^2 + yz = 2^2 \quad \dots(2)$$

$$z^2 + x^2 + zx = (\sqrt{3})^2 \quad \dots(3)$$

and $\text{ar}(\Delta ABC) = \text{ar}(\Delta APB) + \text{ar}(\Delta BPC) + \text{ar}(\Delta APC)$

$$\Rightarrow xy + yz + zx = 2 \quad \dots(4)$$

$$(1) + (2) + (3) \Rightarrow x^2 + y^2 + z^2 = 3$$

$$(1) + (2) \Rightarrow y(x + y + z) = 2$$

$$(2) + (3) \Rightarrow z(x + y + z) = 4$$

$$\text{and } x(x + y + z) = 1$$

$$\therefore x : y : z = 1 : 2 : 4$$

$$\Rightarrow x = \frac{\sqrt{7}}{7}, y = \frac{2\sqrt{7}}{7}, z = \frac{4\sqrt{7}}{7}$$